source ..... make noise · soundhack
rev 2026.10

spectraphon

_travelers guide
_for splitting the light

2 sides	4 modes each	16 arrays per side

› then noon = 12 o'clock tags = the panel drawing

dual spectral oscillator · each side: sam (input drives harmonics) or sao (plays stored arrays) · noise and chaos modes · by tom erbe

01 what it is

sp_01

two sides	A and B, mirror images · each is an oscillator in its own mode
a spectrum	a set of harmonic levels · not a recording · plays at any pitch
like a complex osc	`freq`, `partials`, fm bus, follow/sync and `sine` work the same in every mode

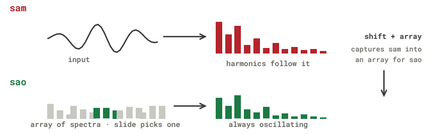
02 modes

sp_02 · fig.01 · mode button

sam · off	the input sets harmonic levels · silent without input
sao · lit	always oscillates · plays spectra from a stored array
noise · chaos	press on past sao (fw sp52+) · the centre LEDs show which: noise 01, chaos 10
mix them	each side has its own mode · fm one with the other either way

__fig.01 : sam and sao

mode off · mode lit



10 first flight

sp_10 · first sound

1. side A: mode off (sam)
2. a voice or drum loop into `a-in` · knob to the tick
3. `partials` full · listen to `odd`
4. turn `slide` then `focus`
5. play `1v/oct` like any vco
6. `[ shift ]` + `[ array ]` to capture, then press mode for sao

03 sam

sp_03 · spectral amplitude modulation

`a-in`	input level · ticks mark modular and line level
`slide`	where the input's first harmonic sits · lower = brighter output
`focus`	band width · low = sharp, quick · high = rings, drones, even echo
`sub/cv`	envelope follower of the input · try it into `slide` or `focus`

04 sao

sp_04 · spectral array oscillation

`slide`	picks the spectrum: first (ccw) to last (cw) in the array
`focus`	cw = compress (brighter, even) · ccw = expand (darker, punchier)
`[ shift ]` 2 s	planar style: focus = fine, slide = coarse
`clk`	steps through the array · adds to slide and focus
`a-in`	detunes the even harmonics up to an octave · cv = fm for evens only

05 shared controls

sp_05 · every mode

`freq` + fine	core pitch with `1v/oct` · `sine` always plays it, even under fm
`partials`	silent → low → high harmonics · alternates odd and even · a timbre vca
`odd` `even`	unpatch even and odd carries both
`sub/cv`	sao: sub osc (A saw, B saturated sine)

06 fm bus · follow · sync

sp_06

`index A` `index B`	internal fm from the other side · both ways at once, no feedback
`[ follow/sync ]`	lit = follow: B pitch becomes an offset from A (noon = 0)
flashing	sync: follow + B hard-synced to A · strong sweeps
beacon	green = octave, fifth, fourth · red = third, sixth
tip	turn follow on before tuning B

07 shift layer

sp_07 · hold shift

`[ shift ]` + `[ array ]`	sam: start / stop array capture · other modes: low frequency (shift stays lit)
`[ shift ]` + `[ cv ]`	sub/cv: sub · stepped random · smooth random · triangle · ramp
`[ shift ]` + other shift	pick array 0 – 15 for this side (binary on the centre LEDs)
… hold 2 s	delete that array, back to default
`[ shift ]`	tap = manual clock · tap tempo for the LFOs

08 noise · chaos

sp_08 · fw sp52+

noise	two sines + noise · partials = sideband width · slide lp, focus hp
chaos	each sine gets a partner · focus = ratio · partials = cross-mod · slide = feedback
`a-in`	detunes the pair up to an octave, as in sao

11 capturing arrays

sp_11 · fig.02

before — partials full
fm bus off · partials full cw and unmodulated

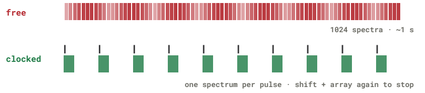
free — no clock 3 s
1024 spectra in about 1 s

clocked — clock in
one spectrum per pulse · shift + array again stops it

tip — keep it moving
static or silent parts become dead spots on slide

__fig.02 : capture

sam · shift + array



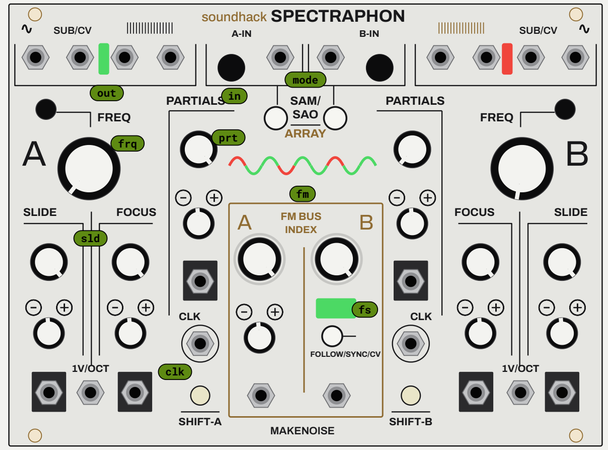
09 specs

sp_09

size	34 hp
power	230 mA +12 V · 55 mA –12 V
arrays	16 per side on microSD · SPECA000.WAV …
firmware	sp_.dat on the card · power on holding mode B

__fig.03 : the panel

numbers match the legend



`out`	**outputs** · sine · sub/cv · odd · even
`in`	**a-in** · **b-in** · input level / even detune
`mode`	**sam/sao/array** · mode · shift + = capture / low freq
`frq`	**freq** · **fine** · core pitch + 1v/oct
`prt`	**partials** · harmonic amount
`sld`	**slide** · **focus** · shape or pick the spectrum
`fm`	**fm bus index** · A ← B · B ← A
`fs`	**follow/sync** · B follows or syncs to A
`clk`	**clk** · **shift** · step array · second layer

12 reset

sp_12

modes	power on holding both `[ shift ]` buttons
one array	delete its file from the card, or hold both shifts 2 s

design homage to the house of x / powers of x data pages — designed by tom muller with jonathan hickman · panel drawing after the make noise panel
not affiliated with make noise, soundhack or marvel